# ANSWER KEY — Integration Lesson 30 Test: Integration by Parts I

For the grader · full working shown · every answer verified by differentiation · give credit for correct method with small arithmetic slips

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**1.** Set up the parts only. (a) `∫x e^{4x} dx` (b) `∫x sin(3x) dx` — state u, dv, du, v and the LIATE rung.

**Work:** LIATE = Logs, Inverse trig, Algebraic, Trig, Exponential; u comes from the highest rung present.
(a) x is **Algebraic**, $e^{4x}$ is Exponential, and A sits above E → u = x, dv = $e^{4x}$ dx. Then du = dx and v = $\int e^{4x}$ dx = (1/4)$e^{4x}$ (divide-by-a rule, a = 4; no +C needed for v).
(b) x is **Algebraic**, sin 3x is Trig, and A sits above T → u = x, dv = sin(3x) dx. Then du = dx and v = ∫sin(3x) dx = −(1/3)cos(3x).
**Answer: (a) u = x (Algebraic), dv = $e^{4x}$ dx, du = dx, v = (1/4)$e^{4x}$. (b) u = x (Algebraic), dv = sin(3x) dx, du = dx, v = −(1/3)cos(3x).**
**Grader note:** the two classic slips here are forgetting the 1/4 and the 1/3 when integrating dv, and losing the minus sign on v in (b) (∫sin = *minus* cos). No credit lost for omitting "+C" on v — that is correct practice.

**2.** Compute `∫(x − 3)e^x dx` and check by differentiating.

**Work:** u = x − 3, dv = $e^x$ dx → du = dx, v = $e^x$.
Assemble uv − ∫v du: (x − 3)$e^x$ − ∫$e^x$ dx = (x − 3)$e^x$ − $e^x$ + C.
Simplify: (x − 3)$e^x$ − $e^x$ = x $e^x$ − 3$e^x$ − $e^x$ = (x − 4)$e^x$.
**Check:** d/dx[(x − 4)$e^x$] = 1·$e^x$ + (x − 4)$e^x$ = $e^x$ + x $e^x$ − 4$e^x$ = (x − 3)$e^x$ ✓ — exactly the integrand.
**Answer: (x − 4)$e^x$ + C (equivalently (x − 3)$e^x$ − $e^x$ + C).**
**Grader note:** both forms are fully correct — do not deduct for leaving it unsimplified. The error to watch for is writing (x − 3)$e^x$ + $e^x$ (sign of the traded integral flipped), which differentiates to (x − 1)$e^x$, not the integrand. Also watch for the student who stops at (x − 3)$e^x$ and never finishes −∫v du.

**3.** Compute `∫(x − 1)cos x dx` and check by differentiating.

**Work:** u = x − 1 (Algebraic beats Trig), dv = cos x dx → du = dx, v = sin x.
Assemble: (x − 1)sin x − ∫sin x dx = (x − 1)sin x − (−cos x) + C = (x − 1)sin x + cos x + C.
**Check:** d/dx[(x − 1)sin x] = sin x + (x − 1)cos x; d/dx[cos x] = −sin x. Total: sin x + (x − 1)cos x − sin x = (x − 1)cos x ✓.
**Answer: (x − 1)sin x + cos x + C.**
**Grader note:** the #1 error is the double negative: ∫sin x dx = −cos x, and *subtracting* it gives **+cos x**. An answer of (x − 1)sin x − cos x + C means one minus sign was dropped; that is the standard sign slip. Half credit if the u/dv choice and assembly are right and only that sign is wrong.

**4.** Compute `∫4x sin x dx` and check by differentiating.

**Work:** linearity first — slide the constant out: $\int 4x \sin x\, dx = 4\int x \sin x\, dx$.
For $\int x \sin x\, dx$: $u = x$, $dv = \sin x\, dx \rightarrow du = dx$, $v = -\cos x$.
Assemble: $x(-\cos x) - \int(-\cos x)\, dx = -x \cos x + \int \cos x\, dx = -x \cos x + \sin x$.
Multiply back by 4: $-4x \cos x + 4 \sin x + C$.
**Check:** $d/dx[-4x \cos x] = -4 \cos x + 4x \sin x$ (product rule); $d/dx[4 \sin x] = 4 \cos x$. Total: $-4 \cos x + 4x \sin x + 4 \cos x = 4x \sin x$ ✓.
**Answer: $-4x \cos x + 4 \sin x + C$.**
**Grader note:** classic errors — (i) multiplying only the first term by 4 ($-4x \cos x + \sin x$); (ii) taking $u = 4x$ and $dv = \sin x\, dx$, which is also perfectly correct ($du = 4\, dx$) and must get full credit if carried out consistently: $4x(-\cos x) - \int(-\cos x)(4\, dx) = -4x \cos x + 4 \sin x$, the same answer; (iii) the forgotten minus in front of $\int v\, du$.

**5.** Why can't u-substitution finish `∫x e^{4x} dx`? Try $u = 4x$; name the rule the integrand came from.

**Work:** u-substitution only undoes the **chain rule**. It needs a "fingerprint": an inside function whose derivative sits nearby as a factor, ready to be absorbed into du.
Try $u = 4x$: then $du = 4\, dx$, so $dx = du/4$, and $x = u/4$. The integral becomes $\int(u/4)e^u(du/4) = (1/16)\int u\, e^u\, du$ — *still* a plain-times-exponential product, exactly the same shape we started with. The substitution only rescaled the problem; it removed nothing.
Try $u = e^{4x}$: $du = 4e^{4x}\, dx$ absorbs the $e^{4x}\, dx$, but the leftover x has to be rewritten as $x = (1/4)\ln u$, giving $\int(1/16)\ln u\, du$ — a by-parts problem again, not simpler.
The real reason: x and $e^{4x}$ are a **product of two unrelated functions** — neither is the derivative of the other's inside, so there is no fingerprint. Nothing here was produced by a chain rule; it has the shape of a **product rule** output. A product needs a product-rule tool: integration by parts, $\int u\, dv = uv - \int v\, du$.
**Answer: no fingerprint exists — neither factor's derivative appears as the other factor, so u-sub has nothing to absorb ($u = 4x$ just rescales to $(1/16)\int u\, e^u\, du$). The integrand came from the product rule, so reverse the product rule: integration by parts.**
**Grader note:** full credit needs both halves — (1) u-sub reverses the chain rule / needs a derivative-fingerprint that is absent, and (2) by parts reverses the product rule. Accept any correct demonstration that a substitution attempt leaves a product behind.

**6.** Compute `∫x e^{4x} dx` and check by differentiating.

> **Work:** $u = x$, $dv = e^{4x}\,dx \to du = dx$, $v = (1/4)e^{4x}$.
> Assemble: $x\cdot(1/4)e^{4x} - \int(1/4)e^{4x}\,dx = (x/4)e^{4x} - (1/4)\cdot(1/4)e^{4x} + C = (x/4)e^{4x} - (1/16)e^{4x} + C$.
> Where the fractions come from: one 1/4 when integrating dv to get v, a second 1/4 when integrating $e^{4x}$ in the traded integral → 1/16.
> **Check:** $d/dx[(x/4)e^{4x}] = (1/4)e^{4x} + (x/4)\cdot 4e^{4x} = (1/4)e^{4x} + x\,e^{4x}$; $d/dx[-(1/16)e^{4x}] = -(1/16)\cdot 4e^{4x} = -(1/4)e^{4x}$. Total: $x\,e^{4x}$ ✓ — the two $(1/4)e^{4x}$ terms cancel.
> **Answer: $(x/4)e^{4x} - (1/16)e^{4x} + C$.**
> **Grader note:** the usual slips are $-(1/4)e^{4x}$ instead of $-(1/16)e^{4x}$ (the second divide-by-4 was skipped) and $-(1/8)e^{4x}$ (adding the 1/4's instead of multiplying). Both are caught instantly by the differentiation check — if the check was not shown, note it.

**7.** Compute `∫x sin(3x) dx` and check by differentiating.

> **Work:** $u = x$, $dv = \sin(3x)\,dx \to du = dx$, $v = -(1/3)\cos(3x)$.
> Assemble: $x\cdot(-(1/3)\cos 3x) - \int(-(1/3)\cos 3x)\,dx = -(x/3)\cos 3x + (1/3)\int\cos 3x\,dx$.
> Finish: $(1/3)\cdot(1/3)\sin 3x = (1/9)\sin 3x$. So the answer is $-(x/3)\cos 3x + (1/9)\sin 3x + C$.
> **Check:** $d/dx[-(x/3)\cos 3x] = -(1/3)\cos 3x + (x/3)\cdot 3\sin 3x = -(1/3)\cos 3x + x\sin 3x$; $d/dx[(1/9)\sin 3x] = (1/9)\cdot 3\cos 3x = (1/3)\cos 3x$. Total: $x\sin 3x$ ✓.
> **Answer: $-(x/3)\cos 3x + (1/9)\sin 3x + C$.**
> **Grader note:** two independent minus signs live here — one inside $v = -(1/3)\cos 3x$, one in front of $\int v\,du$ — and they cancel to give the + on the second term. Answers of $-(x/3)\cos 3x - (1/9)\sin 3x + C$ (double-negative mishandled) or $+(x/3)\cos 3x$ (v's minus dropped) are the two expected wrong answers. Also watch 1/3 vs 1/9.

**8.** On `∫x cos x dx` a student picks $u = \cos x$, $dv = x\,dx$. Carry it one step; (a) false or just unhelpful? (b) the tell? (c) the choice that works.

> **Work:** $u = \cos x$, $dv = x\,dx \rightarrow du = -\sin x\,dx$, $v = x^2/2$.
> Assemble: $\int x \cos x\,dx = (x^2/2)\cos x - \int (x^2/2)(-\sin x)\,dx = (x^2/2)\cos x + \int (x^2/2)\sin x\,dx$.
> (a) The equation is **true** — it is a valid application of $\int u\,dv = uv - \int v\,du$. Nothing illegal happened; it is simply *unhelpful*.
> (b) The tell: the new integral $\int (x^2/2)\sin x\,dx$ contains $\mathbf{x^2}$ where the original had only x — the power went **UP**, so the traded problem is harder than the one we started with. Rule of thumb: if the new integral looks worse, back up and swap u and dv.
> (c) The choice that works is $u = x$, $dv = \cos x\,dx$ (LIATE: Algebraic sits above Trig) $\rightarrow du = dx$, $v = \sin x$, so $\int x \cos x\,dx = x \sin x - \int \sin x\,dx = x \sin x + \cos x + C$.
> **Check:** $d/dx[x \sin x + \cos x] = (\sin x + x \cos x) - \sin x = x \cos x$ ✓.
> **Answer: (a) true, just unhelpful. (b) the power of x rose from x to $x^2$, so the new integral is harder. (c) $u = x$, $dv = \cos x\,dx \rightarrow x \sin x + \cos x + C$.**
> **Grader note:** the key insight being graded is (a) — a bad choice of u is *not* a mathematical error, only a wrong direction. Deduct if the student calls the student's equation "wrong." Also expect $du = +\sin x\,dx$ (missing minus on the derivative of cos x); note that in (c) subtracting $-\cos x$ is what produces the $+\cos x$.

**9.** Evaluate `∫₀¹ x e^{2x} dx` ($\int_0^1 x\, e^{2x}\,dx$) exactly, and say why the answer must be positive.

> **Work:** first the antiderivative. $u = x$, $dv = e^{2x}\,dx \rightarrow du = dx$, $v = (1/2)e^{2x}$.
> $\int x\, e^{2x}\,dx = (x/2)e^{2x} - \int (1/2)e^{2x}\,dx = (x/2)e^{2x} - (1/4)e^{2x}$. Call this F(x).
> **Check F:** $d/dx[(x/2)e^{2x}] = (1/2)e^{2x} + (x/2)\cdot 2e^{2x} = (1/2)e^{2x} + x\, e^{2x}$; $d/dx[-(1/4)e^{2x}] = -(1/2)e^{2x}$. Total $x\, e^{2x}$ ✓.
> Now the FTC: $F(1) = (1/2)e^2 - (1/4)e^2 = (1/4)e^2$.   $F(0) = 0\cdot(1/2)\cdot 1 - (1/4)\cdot 1 = -1/4$.
> Integral $= F(1) - F(0) = (1/4)e^2 - (-1/4) = (e^2 + 1)/4$.
> Sense-check: $e^2 \approx 7.389$, so the value is $\approx 8.389/4 \approx 2.10$. The integrand $x\, e^{2x}$ runs from 0 up to $e^2 \approx 7.4$ across a width-1 interval, so a total near 2 is plausible ✓.
> Why positive: on [0, 1] both $x \geq 0$ and $e^{2x} > 0$, so the integrand is never negative — the whole region sits on or above the axis.
> **Answer: $(e^2 + 1)/4$  ($= e^2/4 + 1/4 \approx 2.10$), positive because $x\, e^{2x} \geq 0$ on [0, 1].**
> **Grader note:** accept $e^2/4 + 1/4$, $(1/4)(e^2 + 1)$, or $(1/4)e^2 + 1/4$ — all identical. Classic errors: $F(0) = 0$ (forgetting that $-(1/4)e^0 = -1/4$, since $e^0 = 1$) which gives $e^2/4$; and subtracting in the wrong order. Decimal-only answers lose the "exact" requirement.

**10.** (a) `∫(2x + 1)e^{−x} dx`, simplified. (b) Check by differentiating. (c) Evaluate $\int_0^1 (2x + 1)e^{-x}\,dx$ exactly.

> **Work (a):** $u = 2x + 1$, $dv = e^{-x}\,dx \to du = 2\,dx$, $v = -e^{-x}$ (since $e^{-x}$ is $e^{ax}$ with $a = -1$, integrating costs a factor $1/(-1)$).
>
> Assemble: $(2x + 1)(-e^{-x}) - \int(-e^{-x})(2\,dx) = -(2x + 1)e^{-x} + 2\int e^{-x}\,dx = -(2x + 1)e^{-x} - 2e^{-x} + C$.
>
> Simplify: factor out $-e^{-x}$: $-(2x + 1 + 2)e^{-x} = -(2x + 3)e^{-x}$.
>
> **(b) Check:** $d/dx[-(2x + 3)e^{-x}] = -2e^{-x} - (2x + 3)(-e^{-x}) = -2e^{-x} + (2x + 3)e^{-x} = (2x + 3 - 2)e^{-x} = (2x + 1)e^{-x}$ ✓.
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> **(c) FTC** with $F(x) = -(2x + 3)e^{-x}$: $F(1) = -(2 + 3)e^{-1} = -5/e$. $F(0) = -(0 + 3)e^{0} = -3$.
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> Integral $= F(1) - F(0) = -5/e - (-3) = 3 - 5/e$.
>
> Sense-check: $5/e \approx 5 \div 2.71828 \approx 1.839$, so the value $\approx 1.16 > 0$. The integrand runs from 1 down to $3/e \approx 1.10$ on $[0, 1]$, all positive, so a total just above 1 is exactly right ✓.
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> **Answer: (a) $-(2x + 3)e^{-x} + C$ (c) $3 - 5/e \approx 1.16$.**
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> **Grader note:** the minus signs pile up three deep here — in v, in front of $\int v\,du$, and in the chain rule during the check. Expected wrong answers: $-(2x + 1)e^{-x} - 2e^{-x}$ left unsimplified (that is CORRECT — give full credit), $-(2x - 1)e^{-x}$ (sign slip when combining), and $5/e - 3$ (endpoints subtracted backwards). Any answer equal to $3 - 5/e$, e.g. $(3e - 5)/e$, is fine.